CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

Chapter 2: Power Play

Topic-wise short notes with solved examples

Exponents are a short way of writing repeated multiplication. This chapter covers the laws of exponents, zero and negative exponents, powers of 10, scientific notation, and how fast exponential growth is.

1 Exponential notation

- n^a means n multiplied by itself a times. Here n is the **base** and a is the **exponent** (or power).
- $5^4 = 5 \times 5 \times 5 \times 5 = 625$ is read "5 raised to the power 4" or "the 4th power of 5".
- Do not confuse $4 + 4 + 4 = 3 \times 4 = 12$ with $4 \times 4 \times 4 = 4^3 = 64$.
- With letters: $a \times a \times a \times b \times b = a^3b^2$.
- **Negative bases**: an even power is positive and an odd power is negative. $(-2)^4 = 16$, $(-4)^3 = -64$, $(-1)^{56} = 1$, $(-1)^5 = -1$.
- $0^n = 0$ for every counting number n.

Example 1.1 Write 7200 as a product of powers of its prime factors.

SOLUTION

$7200 = 72 \times 100 = (2^3 \times 3^2) \times (2^2 \times 5^2)$

$7200 = 2^5 \times 3^2 \times 5^2$

Example 1.2 Find the value of $(-3)^3 \times (-2)^4$.

SOLUTION

$(-3)^3 = -27$ and $(-2)^4 = 16$

-432

2 Exponential growth

- When a quantity is **multiplied** by the same number at every step, it grows exponentially. This is far faster than adding the same amount each time.
- A sheet of paper 0.001 cm thick doubles in thickness with each fold. After k folds it is 0.001×2^k cm thick.

Folds	10	20	30	46
Thickness (about)	1.02 cm	10.5 m	10.7 km	more than 7,00,000 km

- Every 10 extra folds multiply the thickness by $2^{10} = 1024$. After 46 folds the stack would reach beyond the Moon.
- **Linear growth is additive; exponential growth is multiplicative.** A ladder to the Moon (3,84,400 km) with steps 20 cm apart needs about 192 crore steps, but the paper needs only 46 doublings.

Example 2.1 The number of lotuses in a pond doubles every day. The pond is full on day 30. On which day is it half full?

SOLUTION

Since the number doubles each day, the day before the pond is full it is half full.

Day 29

3 Laws of exponents: same base

- **Multiplying**: $n^a \times n^b = n^{a+b}$. Example: $3^4 \times 3^3 = 3^7$.
- **Dividing**: $n^a \div n^b = n^{a-b}$, where $n \neq 0$. Example: $2^{100} \div 2^{25} = 2^{75}$.
- **Power of a power**: $(n^a)^b = (n^b)^a = n^{ab}$. Example: $(4^3)^2 = 4^6$.

Example 3.1 Compute 2^9 using smaller powers.

SOLUTION

$2^9 = 2^4 \times 2^5 = 16 \times 32$

512

Example 3.2 Write 6^{12} as a power of a power in two different ways.

SOLUTION

$12 = 3 \times 4 = 2 \times 6$

$(6^3)^4$ and $(6^2)^6$ (others are possible)

Example 3.3 Simplify $p^4 \times p^6 \div p^7$.

SOLUTION

p^{4+6-7}

p^3

4 Laws of exponents: same exponent

- $m^a \times n^a = (mn)^a$. Example: $2^5 \times 5^5 = 10^5 = 1{,}00{,}000$.
- $\frac{m^a}{n^a} = \left(\frac{m}{n}\right)^a$, where $n \neq 0$. Example: $\frac{10^4}{5^4} = 2^4 = 16$.

Example 4.1 A doubling pond starts with 1 lotus. After 3 days all the lotuses are moved to a pond where the number triples every day. How many are there after 3 more days?

SOLUTION

After 3 days: 2^3. After 3 more days: $2^3 \times 3^3$.

$2^3 \times 3^3 = (2 \times 3)^3 = 6^3$

216

Example 4.2 Are $2^4 \times 3^6$ and $6^4 \times 3^2$ equal?

SOLUTION

$6^4 \times 3^2 = 2^4 \times 3^4 \times 3^2 = 2^4 \times 3^6$

Yes

5 Counting combinations

- If one choice can be made in p ways and a second in q ways, the two together can be made in $p \times q$ ways.
- A code of k places, each with n possible symbols, can be made in n^k ways.

Example 5.1 A lock has 4 slots, each showing a digit from 0 to 9. How many codes are possible?

SOLUTION

Each slot has 10 choices: $10 \times 10 \times 10 \times 10$

$10^4 = 10{,}000$

Example 5.2 A password has 3 places, each a capital letter A–Z or a digit 0–9. How many passwords are possible?

SOLUTION

Each place has $26 + 10 = 36$ choices.

$36^3 = 46{,}656$

6 Zero and negative exponents

- $n^0 = 1$ for every $n \neq 0$, because $n^a \div n^a = n^{a-a} = n^0$ and any non-zero number divided by itself is 1.
- $n^{-a} = \dfrac{1}{n^a}$ and $n^a = \dfrac{1}{n^{-a}}$, where $n \neq 0$. Examples: $2^{-1} = \dfrac{1}{2}$, $10^{-3} = \dfrac{1}{1000}$, $(-5)^{-3} = \dfrac{1}{(-5)^3} = -\dfrac{1}{125}$.
- All the laws of exponents hold when the exponents are any integers.
- n cannot be 0 in these rules, because that would mean dividing by 0.

Example 6.1 Simplify $3^2 \times 3^{-5} \times 3^6$.

SOLUTION

3^{2-5+6}

$3^3 = 27$

Example 6.2 Simplify $2^4 \times (-4)^{-2}$.

SOLUTION

$(-4)^{-2} = \frac{1}{(-4)^2} = \frac{1}{16}$, so the product is $16 \times \frac{1}{16}$.

1

Example 6.3 How many times larger than 4^{-2} is 4^2?

SOLUTION

$4^2 \div 4^{-2} = 4^{2-(-2)} = 4^4$

256 times

7 Powers of 10 and expanded form

- Place values are powers of 10: $1000 = 10^3$, $100 = 10^2$, $10 = 10^1$, $1 = 10^0$, $\frac{1}{10} = 10^{-1}$, $\frac{1}{100} = 10^{-2}$.

Example 7.1 Write 47.305 in expanded form using powers of 10.

SOLUTION

$(4 \times 10^1) + (7 \times 10^0) + (3 \times 10^{-1}) + (0 \times 10^{-2}) + (5 \times 10^{-3})$

Indian system	Power of 10	International system	Power of 10
Lakh	10^5	Million	10^6
Crore	10^7	Billion	10^9
Arab	10^9	Trillion	10^{12}
Kharab	10^{11}	Quadrillion	10^{15}

- 10^{100} is called a googol.

8 Scientific notation

- In **scientific notation** (standard form) a number is written as $x \times 10^y$, where $1 \le x < 10$ and y is an integer.
- $5900 = 5.9 \times 10^3$, $20800 = 2.08 \times 10^4$, $80{,}00{,}000 = 8 \times 10^6$.
- The exponent tells the size of the number and matters most. Changing 2×10^7 to 3×10^7 adds one half; changing it to 2×10^8 makes it 10 times as large.

- To compare numbers in scientific notation, compare the exponents first. If they are equal, compare the coefficients.
- To multiply or divide, work with the coefficients and the powers of 10 separately, then adjust so that $1 \leq x < 10$.

Example 8.1 Write 70,04,00,00,000 in scientific notation.

SOLUTION

The decimal point moves 10 places to the left.

7.004×10^{10}

Example 8.2 There are about 2×10^{16} ants and 8×10^{9} people in the world. About how many ants are there for each person?

SOLUTION

$$\frac{2 \times 10^{16}}{8 \times 10^{9}} = 0.25 \times 10^{7}$$

2.5×10^{6}

Example 8.3 A country has 4×10^{7} households and each uses about 150 litres of water a day. Find the total daily use in scientific notation.

SOLUTION

$4 \times 10^{7} \times 1.5 \times 10^{2} = 6 \times 10^{9}$

6×10^{9} **litres**

9 Estimating with large numbers

- For questions with no exact data: first **guess**, then write the relationship between the quantities, make **reasonable assumptions**, and calculate.
- The aim is the right power of 10, not an exact figure. Different sensible assumptions give different but nearby answers.

Example 9.1 Roughly how many seconds are there in a year? Write the answer in scientific notation.

SOLUTION

$60 \times 60 \times 24 \times 365 = 3{,}15{,}36{,}000$

About 3.15×10^{7} **seconds**

Example 9.2 How old is a person who has lived for 10^9 seconds?

SOLUTION

$10^9 \div (3.15 \times 10^7) \approx 31.7$

About 31.7 years

- 10^6 seconds is under 12 days, but 10^9 seconds is nearly 32 years. Three more powers of 10 make a huge difference.

10 Common mistakes

- Reading 2^3 as 2×3.
- Adding exponents when the bases are different: $2^3 \times 3^2$ is **not** 6^5.
- Multiplying exponents in a product: $n^a \times n^b$ is n^{a+b}, while $(n^a)^b$ is n^{ab}.
- Writing $n^0 = 0$. It is 1.
- Thinking 2^{-3} is negative. It is $\frac{1}{8}$.
- Writing $(-2)^4$ and -2^4 as the same. The first is 16, the second is -16.
- Leaving a coefficient outside 1 to 10 in scientific notation, such as 25×10^5 instead of 2.5×10^6.

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